Question 1:

Question: Which of the following decision(s) is(are) related to managing capital structure? I. Company ABC planning to issue preferred stocks to raise funds II. Company ABC opting to increase its loan size from a financial institution by $10 million III. Company ABC choosing to use retained earnings for purchasing new machinery IV. Company ABC deciding to refinance existing debt at lower interest rates
Type: multiple-choice

Options:

a. I only

b. II only

c. III only

d. IV only

e. I and II only

f. II and IV only

Question 2:

Which of the following statements about different business organizational structures is accurate?
I. In an LLC, members have limited liability protection similar to shareholders in a corporation but with more flexibility.
II. A sole proprietorship faces higher operational risks due to unlimited personal liability for debts and obligations.
III. Corporations can raise capital more easily than partnerships or sole proprietorships because they can issue shares of stock to investors.
IV. Limited partnerships offer pass-through taxation benefits, similar to an S corporation but with different ownership restrictions.
V. A disadvantage of a partnership is that it requires unanimous consent for major business decisions and expansions.

Options:

a. I, II, III and IV only

b. II, III, IV and V only

c. I, III, IV, and V only

d. I, II, III, and IV only

e. II, III, IV, V and a different form of liability structure only

Question 3:

Which of the following statement(s) is(are) true about a premium bond? I. Price of bond will increase over time as it approaches maturity, given all else unchanged II. Before the bond matures, the price of the bond is higher than its face value III. Coupon rate of the bond is higher than the yield to maturity IV. Bond value will decrease if coupon rate decreases, given all else unchanged V. Value of the bond would increase if the yield to maturity increases, given all else unchanged

Type: multiple-choice

Options:

- a. II, III and IV only
- b. I, II, and IV only
- c. I, II, III, and IV only
- d. II, III, IV and V only
- e. I, II, III and V only
- f. All I, II, III, IV and V are true

Question 4:

Which of the following bonds exhibits the least interest rate risk?

Bond X has a maturity of 12 years, an annual coupon payment of $50, and a face value of $1,500. Bond Y will mature in 8 years with a coupon rate of 6%. Bond Z is a zero-coupon bond set to mature in 18 years. Bond W offers an annual coupon of $30 for its entire term of 25 years. Bond V has a maturity period of 14 years, pays out an annual coupon of $360, and has a face value of $2,500.
Type: multiple-choice

Options:

a. Bond X

b. Bond Y

c. Bond Z

d. Bond W

e. Bond V

f. Not enough information is available

Question 5:

Question: FastGrow Credit Union aims to charge an EAR of 12% on their loans. If monthly compounding is applied, what APR should the credit union report? Choose the closest answer.
Type: multiple-choice

Options:

a. 11.5%

b. 11.6%

c. 11.8%

d. 12.0%

e. 12.4%

Question 6:

Given the details about stock XYZ, which pays an annual dividend of $3 today and has a

current price of $240, and assuming one year from now (after Year-1's dividend is paid) its price will be $252. What are the required return, capital gains yield, and dividend yield for stock XYZ? Choose the closest answer.
Type: multiple-choice

Options:

a. 7%, 5%, 1%

b. 6.5%, 4%, 1.5%

c. 6%, 3%, 2%

d. 5.5%, 2%, 3%

e. 5%, 0%, 4%

Question 7:

Given the table below showing the credit ratings of bonds and their corresponding yields to maturity (YTM), suppose you have a bond with an A rating that matures in 15 years, makes semi-annual coupon payments, and has a current price of $2500. If the YTM for this bond is known to be 8%, what would be the annual coupon payment? Choose the closest answer.

Type: multiple-choice

Options:

a. $160

b. $320

c. $480

d. $640

e. $800

Question 8:

Given the details about Stock Omega, which paid out a dividend of $40 today and plans to increase this payment by 3% annually for the next 20 years, followed by constant payments thereafter, and considering a required return of 8%, alongside Stock Delta that will pay an annual dividend of $10 indefinitely starting from today with no growth in dividends, and assuming Stock Delta's current market price is $560. Which of the following statements are true regarding Stock Omega and Stock Delta?

I. The required return for Stock Omega is higher than that for Stock Delta when considering their respective growth rates.
II. The price of Stock Omega today is lower than that of Stock Delta.

Options:

a. None of the statements are true

b. I only

c. II only

d. Both I and II are true

Question 9:

Given an investment opportunity costing $400,000 at the start of the year, it promises to pay you $45,000 at the end of each year for a period of 15 years. If this investment is expected to recoup its initial cost over these 15 years, what would be the approximate Annual Percentage Rate (APR) of return? Choose from the following options:

a. 4.8%

b. 5.2%

c. 5.6%

d. 6.0%

e. 6.4%

Question 10:

Which of the following statements about the term structure is accurate? I. If the current term structure is upward sloping, an increase in expected long-term real interest rates compared to short-term real interest rates will steepen the curve. II. When the term structure slopes downward and real interest rates are constant across maturities, it implies that anticipated future inflation is lower than current inflation levels. III. Under conditions where the term structure is sloping downward with flat real rates, long-term interest rate premiums are generally expected to be negative or smaller than short-term premiums.
Type: multiple-choice

Options:

a. I and II only

b. II and III only

c. I, II and III

d. None of the statements are true

Question 11:

Assume today is January 1, 2024. You are planning for your retirement on January 1, 2039, and you have two retirement goals. Goal 1: Six months after you retire, you want to purchase a luxury car worth $95,000 as a lump sum payment at the beginning of the sixth month that you retire. Goal 2: Once you retire, at the end of each month you are going to withdraw some money for daily and medical expenses. Your first withdrawal will be at the end of the month that you retire. For your first withdrawal, you want to take $15,000, and after that, you expect that the expenses are going to grow by 0.3% each month and last for 28 years. APR is 7% compounded monthly. a. To fulfil your two retirement goals, what is the total amount that you need to have saved by the time you retire? b. On average your salary is expected to grow at

a rate of 0.15% each month, and you are going to deposit 35% of your monthly salary into a savings account. If by the time you retire you have saved $Y in total (amount as calculated in part (a)), what is your monthly salary when you make your first deposit? What is your monthly salary when you make your third deposit?
Type: short-answer

Options: No options provided.

Question 12:

A company named XYZ has issued two bonds: Bond X and Bond Y. Bond X is a 10% annual coupon bond with 20 years to maturity, while Bond Y is a semi-annual coupon paying bond with a coupon rate of 8%, also maturing in 20 years but currently trading at a discount of 5% below par value. Both bonds have a face value of $5000 and the current yield to maturity (YTM) for both is 7%.

a. What is the price of Bond X today?

b. Calculate the Macaulay duration of Bond Y.

c. If after 10 years, the YTM drops to 3%, what would be the new prices of Bonds X and Y?

Question 13:

Company XY plans to launch a major technology initiative alongside Partner ZW, with an initial investment of $20 million at the start of this year. The project will span over 5 years and is expected to generate a revenue of $30 million upon completion, paid out in one lump sum two months later. Seventy percent of the project's cost will be borrowed from a financial institution on a 15-year loan term with monthly fixed payments starting at the end of the first month. If the annual percentage rate (APR) is set at 4%, what would be the monthly payment? Additionally, if Company XY decides to settle all remaining debt obligations immediately after receiving Partner ZW's payment but before making that month’s installment, how much will this final lump sum repayment amount to?

Question 14:

Given the details about dividends paid by stock ZY issued by Company ZY, where each dividend was paid on January 1st starting from 2003 with an initial payment of $5, followed by a decrease of $1 per year for three years due to financial preparation for a major expansion. Afterward, no dividends were paid during the period when the expansion took place (from 2006 to 2008). Upon completion of the project in January 2009, a dividend of $3 was issued and the company announced a growth rate of 1% for the next two years, increasing by 1 percentage point each year until it reached 4%, after which the dividends would grow at 4% indefinitely. If the required return on stock ZY is 8%, what would be the price of stock ZY on January 1st, 2003?